Clinic Appointment System

Refined Product Backlog

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1. Product Backlog

The following Product Backlog contains the prioritized user stories for the Clinic Appointment System. Each backlog item includes its priority order, user story ID, related functional or non-functional requirement, user story description, acceptance criteria, and priority classification.

The backlog has been refined to improve clarity, remove ambiguity, establish testable acceptance criteria, and ensure consistency with the updated User and System Requirements.

1.1 Functional User Stories

Order	ID	Requirement Reference	User Story	Acceptance Criteria	Priority
1	US-01	FR-01 – User Registration	As a patient, I want to create an account so that I can access the Clinic Appointment System.	1. Patient can enter the required registration information.	Must Have
				2. System validates the submitted information.	
				3. System prevents registration with incomplete or invalid required information.	
				4. System creates the account when registration is successful.	
2	US-02	FR-02 – Secure Login	As a registered user, I want to log in securely so that I can access the functions available to my role.	1. User can enter login credentials.	Must Have
				2. System validates the credentials.	
				3. System grants access when credentials are valid.	
				4. System rejects invalid credentials and displays an appropriate message.	
3	US-03	FR-03, FR-37 – Role-Based Access Control	As a user, I want the system to provide role-based access so that I can only access functions and information appropriate to my role.	1. System identifies the user's assigned role after login.	Must Have
				2. System provides access to authorized functions.	

				3. System restricts unauthorized functions and information.	
				4. Users cannot access restricted functions by directly navigating to their URLs.	
4	US-15	FR-23 – Doctor Availability Management	As a doctor, I want to manage my availability so that patients can book appointments during my available times.	1. Doctor can set available dates and time slots.	Must Have
				2. Doctor can update their availability.	
				3. System prevents conflicting availability entries.	
				4. Updated availability is reflected in the appointment booking process.	
				5. Existing confirmed appointments are not unintentionally removed when availability changes.	
5	US-05	FR-08 – Doctor Search	As a patient, I want to search for doctors by name or specialty so that I can find a suitable doctor for my appointment.	1. Patient can search using a doctor's name or specialty.	Must Have
				2. System displays doctors matching the search criteria.	
				3. System displays an appropriate message when no matching doctors are found.	
				4. Patient can select a doctor to view their information.	
6	US-06	FR-09, FR-10 – Doctor Information and Availability	As a patient, I want to view a doctor's professional information and availability so that I can choose a suitable doctor and appointment time.	1. Patient can view the doctor's name, specialty, and professional information.	Must Have

				2. Patient can view the doctor's available dates and time slots.	
				3. Displayed information reflects the latest saved doctor information and availability.	
				4. Unavailable or already-booked slots are not displayed as available.	
7	US-07	FR-10, FR-11 – Available Slots and Appointment Booking	As a patient, I want to view available appointment slots and book an appointment so that I can schedule a consultation with a doctor.	1. System displays available appointment dates and time slots.	Must Have
				2. Patient can select a doctor and an available appointment slot.	
				3. System validates the selected slot before completing the booking.	
				4. System saves the appointment successfully.	
				5. The booked slot is no longer available to other patients.	
8	US-08	FR-12, FR-16 – Prevent Double-Booking	As a patient, I want the system to prevent double-booking of appointment slots so that my confirmed appointment remains reserved for me.	1. System checks appointment availability before confirming a booking.	Must Have
				2. System prevents two patients from reserving the same doctor's time slot.	
				3. System handles simultaneous booking attempts without creating duplicate appointments.	
				4. System displays an appropriate message when a selected slot is no longer available.	

9	US-11	FR-17, FR-18 – Appointment Confirmation	As a patient, I want to receive appointment confirmation so that I know my appointment has been successfully scheduled.	1. System generates confirmation after a successful booking.	Must Have
				2. Confirmation includes the doctor's name, appointment date, and time.	
				3. Confirmation includes a unique appointment reference.	
				4. Patient can view the confirmation details.	
				5. Confirmation is not generated for unsuccessful bookings.	
10	US-16	FR-24, FR-25, FR-26 – Doctor Appointment Management	As a doctor, I want to view and manage my appointment schedule so that my appointments remain organized and up to date.	1. Doctor can view their scheduled appointments.	Must Have
				2. Doctor can view relevant appointment details.	
				3. Doctor can access their previous appointment records.	
				4. Doctor can perform permitted schedule management actions.	
				5. Doctor cannot access another doctor's restricted appointment records.	
11	US-04	FR-06, FR-07 – Patient Profile Management	As a patient, I want to manage my profile so that my personal information remains accurate and up to date.	1. Patient can view their profile information.	Should Have
				2. Patient can update permitted personal information.	
				3. System validates updated information.	

				4. System saves valid changes successfully.	
				5. Updated information is displayed when the patient views their profile again.	
12	US-09	FR-13, FR-15 – Appointment Cancellation	As a patient, I want to cancel my appointment so that I can inform the clinic when I can no longer attend.	1. Patient can view appointments eligible for cancellation.	Should Have
				2. Patient can select an appointment to cancel.	
				3. System applies the clinic's cancellation policy.	
				4. System updates the appointment status to Cancelled after successful cancellation.	
				5. The cancelled appointment slot becomes available again.	
				6. System displays a cancellation confirmation.	
13	US-10	FR-14, FR-15, FR-16 – Appointment Rescheduling	As a patient, I want to reschedule my appointment so that I can choose another available time when my plans change.	1. Patient can select an existing appointment to reschedule.	Should Have
				2. System displays alternative available appointment slots.	
				3. System validates the selected alternative slot.	
				4. System updates the appointment after successful rescheduling.	
				5. The original slot becomes available again.	
				6. Patient receives the updated appointment details.	

14	US-12	FR-19, FR-20 – Appointment History	As a patient, I want to view my upcoming and previous appointments so that I can keep track of my appointments.	1. Patient can view upcoming appointments.	Should Have
				2. Patient can view previous appointments.	
				3. System displays relevant appointment details and statuses.	
				4. Appointment information is accurate and up to date.	
				5. Patients can only access their own appointment history.	
15	US-13	FR-21 – Appointment Notifications and Reminders	As a patient, I want to receive appointment reminders so that I do not forget my scheduled appointment.	1. System identifies upcoming appointments.	Should Have
				2. System generates reminders according to the configured reminder schedule.	
				3. Reminder includes relevant appointment details.	
				4. Reminder is delivered through the supported notification channel.	
				5. Cancelled appointments do not generate unnecessary reminders.	
16	US-14	FR-22 – Doctor Profile Management	As a doctor, I want to manage my professional profile so that patients can see accurate information about me.	1. Doctor can view their professional profile.	Should Have
				2. Doctor can update permitted professional information.	
				3. System validates the updated information.	
				4. System saves valid changes successfully.	

				5. Updated information is reflected when patients view the doctor's profile.	
17	US-17	FR-27, FR-31 – Patient Information Management	As a receptionist or administrator, I want to manage patient information so that patient records remain accurate and up to date.	1. Authorized staff can view patient information.	Should Have
				2. Authorized staff can update permitted patient information.	
				3. System restricts access to unauthorized or sensitive information.	
				4. System validates updated information.	
				5. System saves valid changes successfully.	
18	US-18	FR-28, FR-32, FR-34 – Doctor Information Management	As an authorized staff member, I want to manage doctor information so that the clinic has accurate and up-to-date doctor records.	1. Authorized staff can view doctor information.	Should Have
				2. Authorized staff can update permitted doctor information.	
				3. Authorized administrators can manage doctor records and availability.	
				4. System validates updated information.	
				5. System saves valid changes successfully.	
19	US-19	FR-29, FR-30, FR-35 – Appointment Management	As a receptionist or administrator, I want to manage appointments so that I can support the clinic's appointment scheduling process.	1. Authorized staff can view scheduled appointments.	Should Have
				2. Receptionists can create appointments on behalf of patients.	

				3. Authorized staff can cancel or reschedule appointments according to their permissions and clinic policies.	
				4. System validates appointment availability before confirming changes.	
				5. System updates appointment records successfully.	
20	US-20	FR-36 – Clinic Information Management	As an administrator, I want to manage clinic information so that the system contains accurate and up-to-date clinic details.	1. Authorized administrators can view clinic information.	Could Have
				2. Authorized administrators can update permitted clinic information.	
				3. System validates updated information.	
				4. System saves valid changes successfully.	
21	US-21	FR-04 – User Logout	As an authenticated user, I want to log out of my account so that my account remains protected when I finish using the system.	1. User can select the logout option.	Must Have
				2. System terminates the authenticated session.	
				3. System redirects the user to the login page or another appropriate public page.	
				4. The user cannot access protected pages using the terminated session.	
22	US-22	FR-05, FR-33 – Staff Account Management	As an administrator, I want to create and manage doctor and receptionist accounts so that authorized clinic staff can access the system.	1. Administrator can create doctor and receptionist accounts.	Should Have

				2. Administrator can update permitted account information.	
				3. Administrator can manage account status according to system permissions.	
				4. System assigns the appropriate role to each account.	
				5. Unauthorized users cannot create or manage staff accounts.	

1.2 Non-Functional User Stories

Because your refined System Requirements also contain non-functional requirements, they should be represented in the Product Backlog where the professor requires NFR user stories.

These stories cover the quality attributes of the system rather than individual business features.

Order	ID	Requirement Reference	User Story	Acceptance Criteria	Priority
23	US-23	NFR-01 to NFR-06 – Security	As a user, I want my personal information and account to be protected so that my information cannot be accessed by unauthorized users.	1. User passwords are stored securely.	Must Have
				2. Authentication is required to access protected functions.	
				3. Role-based permissions are enforced.	
				4. Unauthorized access to restricted information is prevented.	
24	US-24	NFR-07 to NFR-10 – Usability	As a user, I want a simple and understandable interface so that I can use the system's functions easily.	1. Main functions are easy to locate.	Should Have
				2. Navigation is consistent across system modules.	

				3. Successful operations display confirmation messages.	
				4. Invalid inputs display understandable error messages.	
25	US-25	NFR-11, NFR-12 – Performance	As a system user, I want the system to respond efficiently so that I can complete appointment-related tasks without unnecessary delays.	1. Normal operations meet the agreed response-time target.	Should Have
				2. The system supports the agreed number of simultaneous users.	
				3. Performance is evaluated under normal operating conditions.	
26	US-26	NFR-13 to NFR-15 – Reliability and Data Consistency	As a clinic user, I want the system to maintain accurate and consistent appointment information so that appointment records remain reliable.	1. Appointment records are stored accurately.	Must Have
				2. Duplicate appointment records are prevented.	
				3. Simultaneous operations do not create inconsistent appointment data.	
				4. Appointment information remains consistent after booking, cancellation, and rescheduling.	
27	US-27	NFR-16 – Data Backup	As an administrator, I want important system data to be backed up so that information can be protected against accidental loss.	1. The system provides a backup mechanism.	Could Have
				2. Important system data is included in backups.	
				3. Backups can be accessed by authorized personnel for recovery purposes.	

28	US-28	NFR-17 to NFR-20 – Maintainability and Compatibility	As a development and maintenance team member, I want the system to be modular, documented, and compatible with common browsers and screen sizes so that it can be maintained and accessed easily.	1. The system follows a modular structure.	Should Have
				2. Technical documentation is available.	
				3. The system works on commonly used modern browsers.	
				4. The interface adapts to supported desktop, tablet, and mobile screen sizes.	